

Block 3 pre-registered predictions vs observed outcomes (stretch testcase)

audit anchor: 645626e (logged 2026-05-19, before any singlezone_commercial_hydronic run)

Prediction	a-priori P	Pre-registered range	Observed	Verdict
mode=none controller verdict	0.80	FAIL	PASS (m_s = 0.431)	FALSIFIED
mode=full surrogate RMSE_T gain	0.70	[50, 90]%	87.81%	CONFIRMED
C_zon ratio — hypothesis A (uniform)	0.35	[1.7, 2.2]×	1.909×	CONFIRMED
C_zon ratio — hypothesis B (scale-dep)	0.50	[3, 10]×	1.909× (outside range)	FALSIFIED
C_zon ratio — hypothesis C (pipeline failure)	0.15	Stage A/B/C fails	all 3 converged	FALSIFIED

*2 surprises shifted support to LOWER a-priori alternatives (Popperian falsifiability):
(i) controller PASS at stretch (was 0.20 prior) | (ii) C_zon uniform across hydronic family (was 0.35 prior)*